

NETWORKS AND COMPLEXITY

Solution 23-5

This is an example solution from the forthcoming book Networks and Complexity.

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 23.5: Paper supply [Lvl. 3]

In an office the copier paper is stored in a cabinet. At an average rate of once a week an employee takes paper out of the cabinet and loads it into the copier. There is a 50% chance that the employee tries to load only one packet of paper, and a 50% chance that the employee tries to load two packets of paper. If the cabinet is empty, the office manager immediately orders 4 new packets of paper which arrive on average within 2 weeks.

- a) Consider this system as a network of states, numbered 0 to 4. The system is in state i if there are i packets of paper in the cabinet. Write a system of differential equations for the variables $x_0, \dots, x_4$, where x_i is the probability that the system is in state i .

Solution

The system of equations is

$$\dot{x}_0 = -0.5x_0 + x_1 + 0.5x_2 \quad (1)$$

$$\dot{x}_1 = -x_1 + 0.5x_2 + 0.5x_3 \quad (2)$$

$$\dot{x}_2 = -x_2 + 0.5x_3 + 0.5x_4 \quad (3)$$

$$\dot{x}_3 = -x_3 + 0.5x_4 \quad (4)$$

$$\dot{x}_4 = -x_4 + 0.5x_0 \quad (5)$$

- b) Write the differential equation system in the form $\dot{\mathbf{x}} = -\mathbf{L}\mathbf{x}$ where $\mathbf{x} = (x_0, \dots, x_4)^T$.

Solution

We find

$$\mathbf{L} = \begin{pmatrix} 0.5 & -1 & -0.5 & 0 & 0 \\ 0 & 1 & -0.5 & -0.5 & 0 \\ 0 & 0 & 1 & -0.5 & -0.5 \\ 0 & 0 & 0 & 1 & -0.5 \\ -0.5 & 0 & 0 & 0 & 1 \end{pmatrix} \quad (6)$$

- c) Use Kirchhoff's theorem to compute the probability that there isn't any paper in the cabinet. (i.e. compute x_0 in the steady state).

Solution

The easiest solution is to use Kirchhoff together with the matrix-tree theorem. This yields

$$S_1 = \begin{vmatrix} 1 & -0.5 & -0.5 & 0 \\ 0 & 1 & -0.5 & -0.5 \\ 0 & 0 & 1 & -0.5 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 1 \quad (7)$$

where we have used that the determinant is the product of the eigenvalues, and in this case we can see immediately that all the eigenvalues are 1 due to the upper triangular form of the matrix.

$$S_2 = \begin{vmatrix} 0.5 & -0.5 & 0 & 0 \\ 0 & 1 & -0.5 & -0.5 \\ 0 & 0 & 1 & -0.5 \\ -0.5 & 0 & 0 & 1 \end{vmatrix} \quad (8)$$

$$= 0.5 \begin{vmatrix} 1 & -0.5 & -0.5 \\ 0 & 1 & -0.5 \\ 0 & 0 & 1 \end{vmatrix} + 0.5 \begin{vmatrix} -0.5 & 0 & 0 \\ 1 & -0.5 & -0.5 \\ 0 & 1 & -0.5 \end{vmatrix} = 0.3125 \quad (9)$$

$$S_3 = \begin{vmatrix} 0.5 & -1 & 0 & 0 \\ 0 & 1 & -0.5 & 0 \\ 0 & 0 & 1 & -0.5 \\ -0.5 & 0 & 0 & 1 \end{vmatrix} = 0.375 \quad (10)$$

$$S_4 = \begin{vmatrix} 0.5 & -1 & -0.5 & 0 \\ 0 & 1 & -0.5 & 0 \\ 0 & 0 & 1 & -0.5 \\ -0.5 & 0 & 0 & 1 \end{vmatrix} = 0.25 \quad (11)$$

$$S_5 = \begin{vmatrix} 0.5 & -1 & -0.5 & 0 \\ 0 & 1 & -0.5 & -0.5 \\ 0 & 0 & 1 & -0.5 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 0.5 \quad (12)$$

In summary we have

$$S_1 = \frac{16}{16} \quad S_2 = \frac{5}{16} \quad S_3 = \frac{6}{16} \quad S_4 = \frac{4}{16} \quad S_5 = \frac{8}{16} \quad (13)$$

hence

$$\sum S_i = \frac{39}{16} \quad (14)$$

so we can find the steady state

$$X_1^* = \frac{16}{39} \quad X_2^* = \frac{5}{39} \quad X_3^* = \frac{6}{39} \quad X_4^* = \frac{4}{39} \quad X_5^* = \frac{8}{39}. \quad (15)$$